

Bambu Lab LED Status Bar

(DIY version using Home Assistant + ESP32 + WLED or MQTT)

Please check the Wiring guide available on:

<https://makerworld.com/en/models/2172105-bambu-lab-p2s-led-progress-status-bar>

Home Assistant Version

- Uses Home Assistant + the Bambu Lab integration
- ESP32 running WLED
- Full printer-state and print-progress automation
- Improved HMS / print error handling
- See the attached **Home Assistant Setup Guide V2**

Requirements:

Before starting, you should have:

- Home Assistant running ([Home assistant page](#))
- Bambu Lab integration installed (**more info** [here](#))
- WLED integration installed ([WLED page](#) | [HA Integration](#))

Once those are in place, everything else is mainly wiring, flashing WLED, creating the presets and importing the Home Assistant automation.

Printer Compatibility

P2S	 Tested
X1C	 Community tested
P1S	 Expected to work
Other Bambu models	 Not confirmed yet

I originally built and tested this on my **Bambu Lab P2S**, and the **X1C Home Assistant setup** has **also been confirmed working by other users**.

The P1S should also be compatible, but mounting, cable routing and some Home Assistant entities may be a little different.

If you try this on another Bambu printer, let me know in the comments. it would be great to keep adding confirmed models here.

1. WLED basic configuration

I'm assuming you already flashed WLED onto the ESP32 and it's online in your network. [Flash Web tool](#)

Now we just need to set the LED count and the data pin so the strip behaves like a 10-step progress bar.

1. open WLED in your browser
2. go to Config → LED Preferences
3. set LED count: 10

Scroll down to LED Outputs and confirm the data pin:

1. GPIO: 16 (if you used another pin, adjust here — in my build I used GPIO16)
2. save & reboot WLED

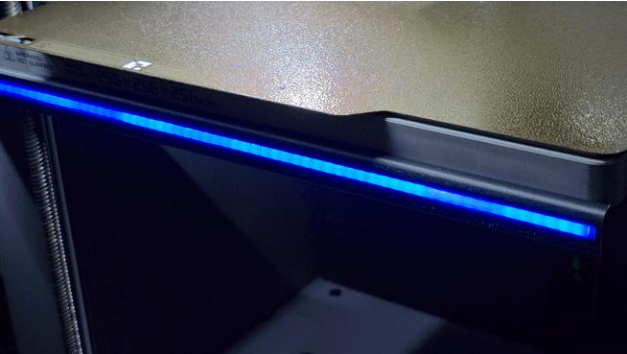
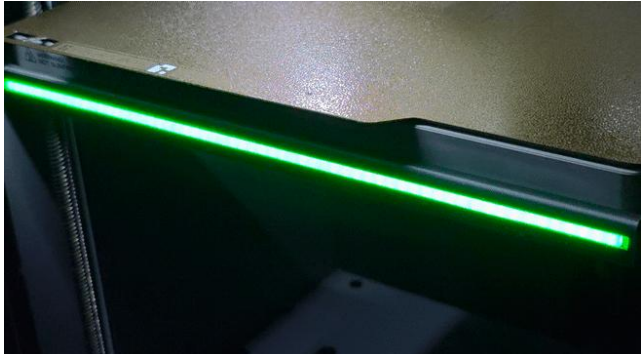
After reboot, the strip should light up normally when you pick any effect.

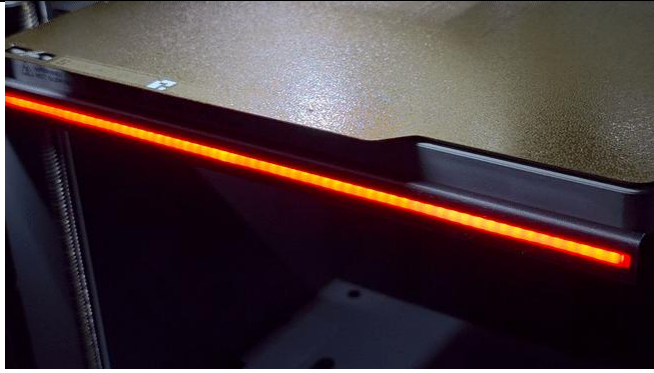
if colors look swapped, change Color Order between RGB/GRB.

4. WLED presets for printer states

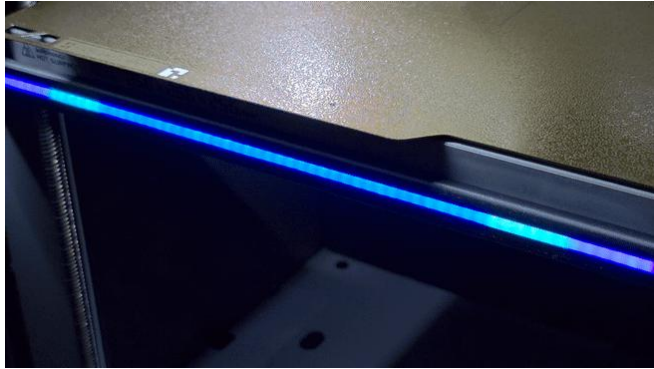
To make the LEDs change depending on what the printer is doing, I created a few WLED presets. These don't have to be fancy, the important part is that the names stay consistent, because Home Assistant will trigger them later.

You can use whatever effects you like, here are the ones I picked:

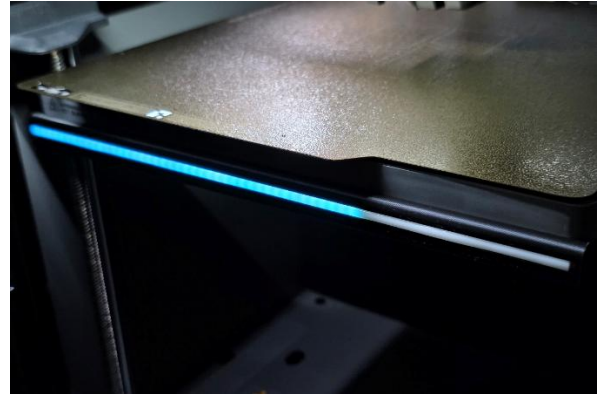
1: P2S-Idle (Plasmoid, Blue & Light Blue) 	2: P2S-Finished (Fade, Green & Black) 
3: P2S-Error (Fade, Red & Black)	4: P2S-Heating (Loading, Red & Orange)



5: P2S-Calibrating (Gravfreq, Light blue & Purple)



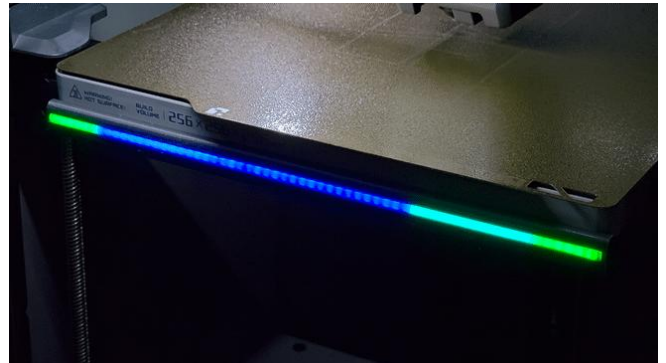
6: P2S-Printing (Percent, White)



7: P2S-Sleep (Plasmoid, Light Blue & black)



8: P2S-Filament (Chase, Green & Blue)



Before setting up Home Assistant, make sure the required WLED presets exist.

The automation uses these exact preset names:

- P2S-Idle
- P2S-Finished
- P2S-Error
- P2S-Heating
- P2S-Calibrating
- P2S-Filament
- P2S-Printing

- P2S-Sleep

You can create them manually or import the included:

WLED-presets-json.txt

Using the included file is the easiest option and gives you the same effects and colours used in my setup.

Importing the Presets

1. Open <http://YOUR-WLED-IP/edit>
2. Back up your existing presets.json if you already use presets
3. Open WLED-presets-json.txt
4. Copy its contents into presets.json
5. Save and restart WLED
6. Confirm the P2S presets appear in WLED

Important: replacing presets.json will overwrite your existing presets, so back it up first if needed.

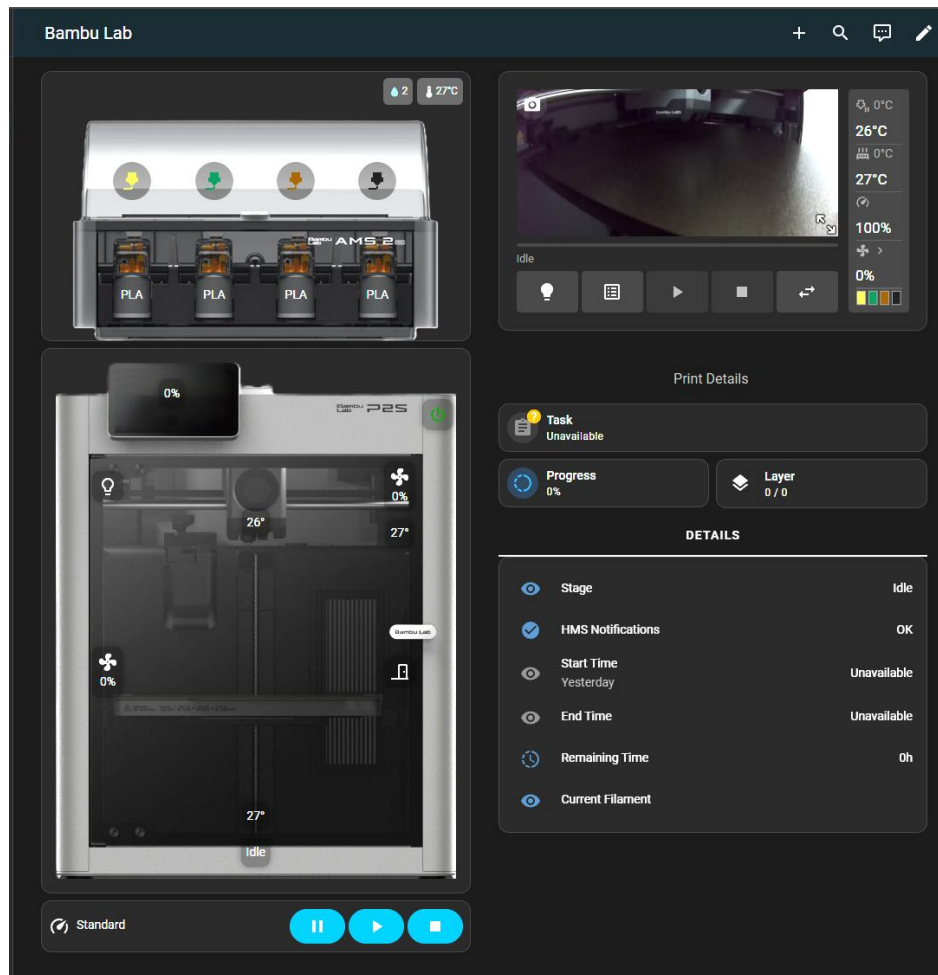
The preset names are case-sensitive and must stay exactly as listed above.

5. Home Assistant automation

With the presets already created in WLED, the last step is to make Home Assistant switch between them automatically.

The LED strip will react based on the preset names you made earlier, so heating, calibrating, printing, finished, etc. all show different effects, and while printing, the strip grows like a 10-step progress bar.

Make sure that the Bambu Lab Integration is installed. [Integration page](#)



How the automation works:

- Home Assistant checks your Bambu printer status
- based on that status, it tells WLED to load the matching preset
- if the printer is printing, the automation also updates intensity every 10% so the strip grows step-by-step
(one “LED” for each 10% because we set LED count to 10 earlier)

Everything is done inside one automation, so it’s easy to maintain and change later.

←

Bambu Lab LED Status Bar V2

Traces

Controls the WLED status bar based on Bambu Lab printer status, current stage, errors and print progress.

When

Print started

P2S_22E8BJ581900934

=

⋮

Print finished

P2S_22E8BJ581900934

=

⋮

Print error

P2S_22E8BJ581900934

=

⋮

Print failed

P2S_22E8BJ581900934

=

⋮

HMS error

P2S_22E8BJ581900934

=

⋮

When P2S_22E8BJ581900934 HMS errors changes to Problem

=

⋮

When P2S_22E8BJ581900934 Print error changes to Problem

=

⋮

When P2S_22E8BJ581900934 HMS errors changes from Problem to OK

=

⋮

When P2S_22E8BJ581900934 Print error changes from Problem to OK

=

⋮

Trigger every 10 seconds of every minute of every hour

Triggered

=

⋮

When P2S_22E8BJ581900934 Print progress changes state or any attributes

=

⋮

When P2S_22E8BJ581900934 Current stage changes to Heating hotend, Heating chamber or Heatbed preheating

=

⋮

When P2S_22E8BJ581900934 Current stage changes to Homing toolhead, Auto bed leveling, Sweeping XY mech mode, Calibrating motor noise or Calibrating extrusion

=

⋮

When P2S_22E8BJ581900934 Current stage changes to Changing filament, Filament loading or Filament unloading

=

⋮

When P2S_22E8BJ581900934 Door changes from Closed to Open

=

⋮

When P2S_22E8BJ581900934 Chamber light changes from On to Off

=

⋮

When P2S_22E8BJ581900934 Chamber light changes from Off to On

=

⋮

And if (optional)

All conditions added here need to be satisfied for the automation to run. A condition can be satisfied or not at any given time, for example: 'If Bruno H is home'. You can use building blocks to create more complex conditions.

+ Add condition

Then do

Choose between 11 options

=

⋮

Option 1: Error detected

=

⋮

Option 2: Recheck active error

=

⋮

Option 3: Print started

=

⋮

Option 4: Update print progress

=

⋮

Option 5: Heating

=

⋮

Option 6: Calibration

=

⋮

Option 7: Filament

=

⋮

Option 8: Print finished

=

⋮

Option 9: Finished to idle when door opens

=

⋮

Option 10: Sleep

=

⋮

Option 11: Awake

=

⋮

+ Add option

+ Add default actions

6. Adding the automation using the YAML file

The full Home Assistant V2 automation is included in the project files as:

BambuLabLEDStatusBar-automation.txt

To install it:

- Open **Home Assistant**
- Go to **Settings → Automations & Scenes → Automations**
- Click **Add Automation → Create New Automation**
- Open the three-dot menu in the top-right corner
- Select **Edit in YAML**
- Paste the contents of **BambuLabLEDStatusBar-automation.txt**

Replace the Placeholders

Before saving the automation, replace the placeholders with the entity IDs from your own Home Assistant setup.

The YAML uses the following placeholders:

- YOUR_BAMBU_DEVICE_ID
- YOUR_PRINTER_print_progress
- YOUR_PRINTER_current_stage
- YOUR_PRINTER_hms_errors
- YOUR_PRINTER_print_error
- YOUR_PRINTER_enclosure_door
- YOUR_PRINTER_chamber_light
- YOUR_WLED_LIGHT
- YOUR_WLED_PRESET
- YOUR_WLED_INTENSITY

You can use **Find / Replace** in your text editor to make sure every occurrence is updated.

Example

Your entities may look something like this:

sensor.p2s_22e8bj581900934_print_progress

sensor.p2s_22e8bj581900934_current_stage

binary_sensor.p2s_22e8bj581900934_hms_errors

binary_sensor.p2s_22e8bj581900934_print_error

binary_sensor.p2s_22e8bj581900934_enclosure_door

light.p2s_22e8bj581900934_chamber_light

light.wled

select.wled_preset

number.wled_intensity

The exact entity names will be different on your Home Assistant installation.

Important

Do not copy the example entity IDs directly.

Make sure you replace the placeholders with the entities belonging to:

- your Bambu Lab printer
- your WLED controller

The WLED preset names must also match the names created earlier in this guide exactly:

- P2S-Idle
- P2S-Finished
- P2S-Error
- P2S-Heating
- P2S-Calibrating
- P2S-Filament
- P2S-Printing
- P2S-Sleep

Preset names are case-sensitive.

Once all placeholders have been replaced:

- Save the automation
- Reload automations if required

Once saved, the LED strip should start reacting automatically to the printer state and print progress.

Known Issue / Note:

When I power the printer through a smart outlet and turn it off, the LED strip sometimes flashes , even though everything should be off. I'm still looking into why it happens, feels like a small amount of leftover voltage or leakage somewhere, but not confirmed yet.

Project Links

For the latest version of the project, updates, files and community feedback, visit the MakerWorld project page:

Bambu Lab P2S LED Progress / Status Bar

[MakerWorld Project Page](#)

If you build this project, adapt it to another Bambu Lab printer, or improve the automation, feel free to share your results in the comments.

I'll keep updating the project page as I make improvements, test more printers and finish the standalone MQTT version.

CAD / Mount Source

You can also modify the LED holder using the original Onshape design:

[Onshape Document](#)

Thanks for checking out the project, and have fun building your own LED status bar. 😊